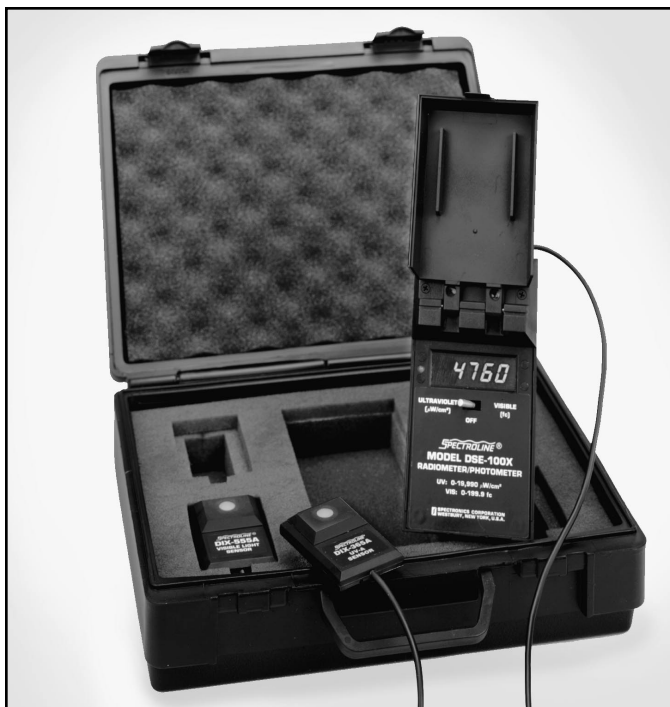

OPERATOR'S MANUAL

SPECTROLINE®

DSE-100 Series Digital Radiometer/Photometers



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Congratulations on choosing a Spectroline® DSE-100 Series digital radiometer/photometer. PLEASE READ THE INSTRUCTIONS IN THIS MANUAL CAREFULLY BEFORE USING THE UNIT.

SECTION I INTRODUCTION

1.1 GENERAL

The Spectroline® DSE-100X digital radiometer/photometer, through the use of two **interchangeable sensor heads**, provides black light (long wave UV) readings from 0-19,990 $\mu\text{W}/\text{cm}^2$, and visible light readings from 0-199.9 fc (0-1999 lux with the DSE-100X/L). For high-range readings, choose the DSE-100H or DSE-100H/L readout unit with the proper sensor. See Table 1-1 for specifications. All models have an overall accuracy of better than $\pm 5\%$ and conform to black light and visible light MIL and ASTM specs for FPI and MPI in NDT applications.

The DIX-365A long wave UV sensor is calibrated to NIST standards by pyroelectrical methods for maximum accuracy. It uses a high quality interference filter that exhibits accurate and carefully controlled spectral coverage, practically eliminating meter sensitivity to infrared and other unwanted radiations. This allows the readings to be obtained directly instead of by calculation. The interference filter also minimizes the "solarization" effect upon the response of the meter over time. The DIX-365A is used with the DSE-100X/L and DSE-100H/L as well as with the DSE-100X and DSE-100H readout units.

The DIX-555A and DIX-555A/L visible light sensors are calibrated photometrically with a quartz-halogen lamp and have a spectral response that closely follows the CIE relative photopic luminosity curve. The DIX-555A is used with the DSE-100X and DSE-100H readout units. The DIX-555A/L is used with the DSE-100X/L and DSE-100H/L readout units.

The sensors are loaded with a low electrical impedance, making their linearity vastly superior to that of any other comparably priced radiometers. The sensors are provided with a special diffuser-sensor window that helps prevent filter degradation and ensures accurate lambertian or cosine response.

The metering electronics of the DSE-100 Series radiometers feature 0.5% accuracy, integrated circuitry, and solid-state LED display. Additional features of the electronics are excellent temperature and over-ranging characteristics. All models have built-in autozeroing.

1.2 FEATURES

1.2.1 The radiometer features nonrechargeable and rechargeable options. With the nonrechargeable option, the meter requires four "AA" 1.5 volt, alkaline batteries. With the rechargeable option, the meter requires four "AA" 1.25 volt, nickel-cadmium batteries and the Spectroline model DM-15C0001 battery charger (separately available).

1.2.2 The radiometer measures irradiances as specified in Table 1-1. The sensors generate electrical current proportional to the irradiance experienced at a plane defined by the face of the sensor. Amplification and current-to-voltage conversion circuitry are contained within the interchangeable sensor heads, and the hand-held display unit provides high reliability and accuracy in the reading.

1.2.3 The numerical display is a light-emitting diode (LED) for precise reading in even total darkness. The numerals are 0.5 inch (12.7mm) high and in a single plane for distant and wide-angle viewing.

1.2.4 All models are equipped with a built-in battery-level indicator light. The green LED is lit when the battery voltage is adequate. The LED turns off when the voltage falls below the minimum acceptable level. The user must then replace or recharge the batteries.

1.3 ACCESSORIES AND REPLACEMENT PARTS

All accessories and replacement parts required for the operation of the radiometer are listed in Table 7-1.

1.4 TECHNICAL DATA

Table 1-1 lists all the technical specifications for the radiometer.

WARNING

This radiometer is carefully designed to prevent accidental shock to the operator when properly used. However, no engineering design can render safe an instrument that is used carelessly. Therefore, recommendations presented in this manual must be read carefully and thoroughly understood before any measurements are made. **Failure to follow directions could result in serious adverse effects.**

The DIX-365A sensor is designed for use in regions of the spectrum, notably the ultraviolet range, that may be hazardous to the eyes and/or skin of individuals. Ultraviolet protective eyewear (UVS-30 spectacles or UVG-50 goggles) and facewear (UVF-80 face shield) are obtainable from Spectronics Corporation for instances when UV exposure is unavoidable.

SECTION II INSTALLATION

2.1 UNPACKING AND INSPECTION

2.1.1 Unpack and inspect the radiometer for possible damage in shipment. Check the performance as soon as possible. If damage is noted, immediately notify the carrier and supplier and do not use the instrument. Also check that all items are included.

2.1.2 Save the shipping carton and packing materials for future storing or shipping of the radiometer.

2.2 POWER SOURCE REQUIREMENTS

The radiometer is battery operated. The batteries are factory installed in the instrument.

2.3 INSTALLATION

The radiometer may be operated in any position without compromising accuracy. The sensor probe may also be positioned at any angle.

TABLE 1-1 TECHNICAL DATA

1. READOUT UNIT				
	DSE-100X	DSE-100X/L	DSE-100H	DSE-100H/L
IRRADIANCE RANGE Long Wave UV Visible Light—Footcandles Visible Light—Lux	0-19,990 $\mu\text{W}/\text{cm}^2$ 0-199.9 fc NA	0-19,990 $\mu\text{W}/\text{cm}^2$ NA 0-1,999 lux	0-199.9 mW/cm ² 0-1,999 fc NA	0-199.9 mW/cm ² NA 0-19,990 lux
RESOLUTION Long Wave UV Visible Light—Footcandles Visible Light—Lux	10 $\mu\text{W}/\text{cm}^2$ 0.1 fc NA	10 $\mu\text{W}/\text{cm}^2$ NA 1 lux	0.1 mW/cm ² 1 fc NA	0.1 mW/cm ² NA 10 lux
DISPLAY Long Wave UV Visible Light—Footcandles Visible Light—Lux	4.5 digit, 7 segment, LED 0.5 in (12.7mm) high 3.5 digit, 7 segment, LED 0.5 in (12.7mm) high NA	4.5 digit, 7 segment, LED 0.5 in (12.7mm) high NA 3.5 digit, 7 segment, LED 0.5 in (12.7mm) high	3.5 digit, 7 segment, LED 0.5 in (12.7mm) high 3.5 digit, 7 segment, LED 0.5 in (12.7mm) high NA	3.5 digit, 7 segment, LED 0.5 in (12.7mm) high NA 4.5 digit, 7 segment, LED 0.5 in (12.7mm) high
CONVERSION RATE	3 readings per second nominal			
OVERALL ACCURACY	Better than $\pm 5\%$ with reference to NIST standards			
TEMP. COEFFICIENT	$\pm 0.025\%/^{\circ}\text{C}$ (10 to 40°C)			
2. SENSOR				
	DIX-365A	DIX-555A (Footcandles)	DIX-555A/L (Lux)	
SPECTRAL RANGE	320-400 nm	380-760 nm	380-760 nm	
TEMP. COEFFICIENT	$\pm 0.20\%/^{\circ}\text{C}$ (10 to 40°C)			
3. POWER REQUIREMENTS—BATTERY OPERATION				
Four nonrechargeable "AA" alkaline battery cells are included as standard.				
Four rechargeable "AA" nicad battery cells and a battery recharger are available as an option.				
4. REFERENCE CONDITIONS—Specified on Certificate of Calibration				
5. DIMENSIONS				
	Length	Thickness	Width	Weight
READOUT UNIT	7.25 in (18.4cm)	2.0 in (5.1cm)	3.5 in (8.9cm)	1 lb (0.45kg)
SENSOR	3.0 in (7.6cm)	0.70 in (1.8cm)	2.0 in (5.1cm)	1.25 oz (35.4gm)
SENSOR CORD	4 ft (1.2m)			

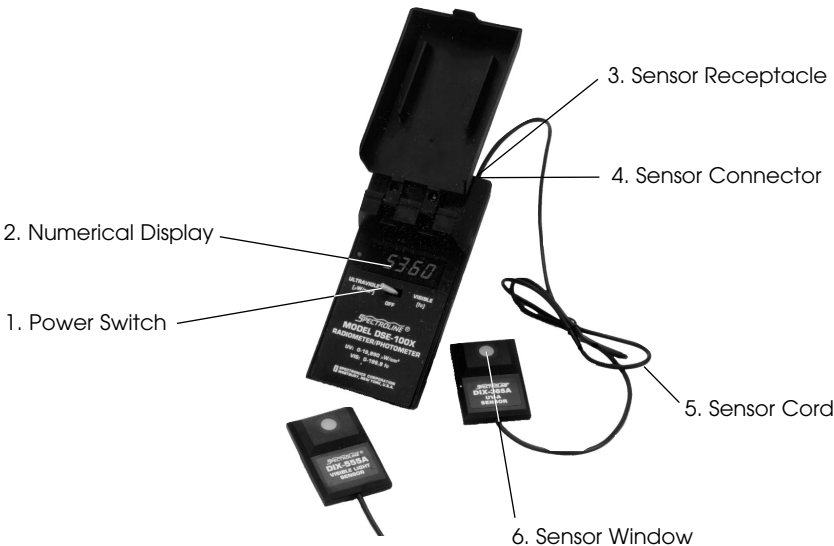
SECTION III CONTROLS, CONNECTORS AND INDICATORS

3.1 GENERAL

All operating controls, connectors and indicators are described in Table 3-1. Become familiar with each item prior to operating the radiometer for the first time.

TABLE 3-1 FRONT AND BACK PANEL DESCRIPTION

1. Power Switch:	With the dual-operation toggle switch flipped to the "ULTRAVIOLET" position, the readout unit will show the long wave UV intensity of the light source being measured, displayed in microwatts per square centimeter ($\mu\text{W}/\text{cm}^2$). With the switch in the "VISIBLE" position, the readout unit will show the visible light emission of the light source, displayed in footcandles (fc) or lux.
2. Numerical Display:	The digital display uses LEDs and includes a "1" digit and 7-segment type 0-9 digits. The last digit is permanently set at "0." Over-range (or out-of-range) condition is indicated by a blank display, except for the most significant "1."
3. Sensor Receptacle:	Accepts sensor connector.
4. Sensor Connector:	On sensor head cord. Insert into sensor receptacle to connect sensor head to readout unit.
5. Sensor Cord:	Connects the signal from the sensor to the readout unit and display circuitry.
6. Sensor Window:	Sapphire diffuser to ensure good cosine response.
7. Battery Level Indicator Light (Not Illustrated)	When lighted, green LED indicates that battery level is adequate.



SECTION IV OPERATION

4.1 GENERAL

This section of the manual contains information required to use and operate the radiometer in a safe and proper manner.

4.2 IRRADIANCE MEASUREMENT PROCEDURE

- a. Be sure that the sensor used is appropriate for the spectral region of interest.
- b. Plug the sensor connector into the receptacle on the back of the readout unit.
- c. Place the sensor at the point where the irradiance/illuminance is to be measured.
- d. Turn on the light source.
- e. Flip the wavelength selector switch to the *left* for ultraviolet or to the *right* for visible light.
- f. The value of the irradiance/illuminance will be expressed in microwatts per square centimeter ($\mu\text{W}/\text{cm}^2$) for ultraviolet with the DSE-100X or DSE-100X/L models, or milliwatts per square centimeter (mW/cm^2) with the DSE-100H or DSE-100H/L models. In addition, visible light will be expressed either in footcandles (fc) with the DSE-100X or DSE-100H models, or in lux with the DSE-100X/L or DSE-100H/L models. See Table 1-1 for specifications.
- g. Return the wavelength selector switch to the OFF (middle) position and remove the sensor from the light source.

NOTES: The least significant digits of the measured irradiance may change as the display is being observed. The amount of variance will depend on the stability of the light source, with the change being smaller for tubular fluorescent-type lamps and larger for HID (high intensity discharge) lamps.

Also, interference filters are not perfect band-pass filters. Therefore, if an inappropriate measurement is attempted (for example, measurement of small amounts of UV-B from an intense UV-A radiation source or measurement of relatively small amounts of UV-A, UV-B or UV-C in sunlight), errors can be expected if appropriate corrections are not made.

Since the radiometer has built-in autozeroing, it does not require any zero adjustment.

CAUTION

This radiometer is designed for **intermittent use only**. **Do not leave the exposed sensor head under the light source any longer than necessary to take measurements.** Prolonged exposure can result in premature aging and/or solarization of the sensor, necessitating more frequent recalibration to maintain accurate readings. In time, such exposure can cause permanent damage to the sensor.

SECTION V THEORY OF OPERATION

5.1 OVERALL SYSTEM

The basic block diagram for the radiometer is shown in Figure 5-1.

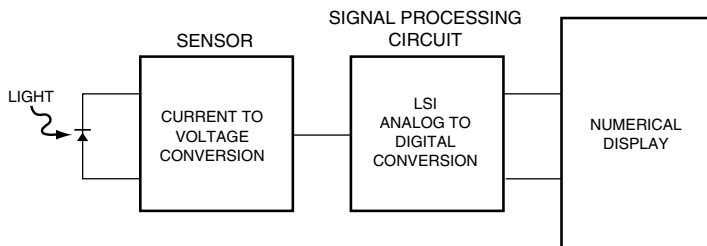


Figure 5-1 Basic System Block Diagram

5.1.1 THE INPUT OPTICS

The lambertian (cosine) response of the sensor head is desirable for many measurement applications, especially those where the angle from the source to the detector is variable or those situations where the angle from the source is "extended," such as in the measurement of a fluorescent lamp at distances comparable to or shorter than its length. In the latter case, the extended source provides radiation from many angles, all of which must be properly "weighted" as to their effectiveness on the plane represented by the sensor surface.

In actual practice, it is difficult to make a sensor conform to the ideal response over the entire solid angle of 2π steradians. The sensor heads used with the readout unit minimize this problem by using a diffusing material (sapphire) having good transmission at 365nm or 555nm, and mounted close to the surface of the sensor so that the oblique rays are not obstructed. The spectral range is selected by providing an appropriate optical filter before the sensor cell assembly.

5.1.2 THE SENSOR CELL

Photovoltaic Operation

When a p-n junction is operated with no externally applied voltage, it is considered to be operating in the photovoltaic mode. Under this zero applied voltage condition and low levels of incident light, the p-n junction will generate a current proportional to the light power incident on the active surface. This photon induced current, or photo-current, will divide between the diode parallel dynamic resistance and the parallel load resistance. The dynamic resistance is normally a high value and is an inverse exponential function of forward voltage. The direction of current flow will produce a voltage across the diode that opposes the band-gap potential of the photodiode junction, thus forward biasing it. As a result, the value of the diode dynamic resistance (R_d) drops exponentially as the irradiance increases and the photogenerated voltage is a quasi-logarithmic function of diode irradiance when the external load resistance is considered. Another major disadvantage is that R_d typically has a wide spread of values over different production batches.

One way of achieving sufficiently low load resistance and eliminating the effect of the diode parallel resistance is to feed the photocurrent into the virtual ground of an operational amplifier.

The output voltage is the result of the photocurrent being driven by the amplifier through the feedback resistor and the input impedance $R_{in} = R_f / A$ where A is the open loop gain and R_f the feedback resistor.

This circuit has a linear response and is low noise due to the almost complete elimination of leakage current with the zero bias. This results in a proportional voltage being presented to the signal conditioning section of the electronics.

5.1.3 ANALOG-TO-DIGITAL CONVERTER SECTION

The analog-to-digital converter uses a large-scale integrated (LSI) circuit and changes the DC voltage into digital information. This digital information is decoded into 7-segment information for the LED display.

5.1.4 DISPLAY

The DSE-100X and DSE-100X/L readout units feature a 4.5 digit LED display that reads UV directly in microwatts per square centimeter ($\mu W/cm^2$). The DSE-100H and DSE-100H/L models have a 3.5 digit display that reads UV in milliwatts per square centimeter (mW/cm^2). Visible light readings (foot candles or lux) for all readout units are displayed in 3.5 digits, except for the DSE-100H/L, which displays in 4.5 digits.

SECTION VI OPERATOR SERVICING

WARNING

The following operator servicing instructions are intended to be used by qualified personnel for limited maintenance of the radiometer. **Do not perform any servicing other than that contained in these operator instructions.** Return the unit to the factory for other maintenance not covered in this section.

6.1 GENERAL

The radiometer is carefully designed and constructed with high quality components. By providing reasonable care and following the instructions in this manual, the user can expect a long, useful service life from the instrument.

6.2 WARRANTY

The warranty policy for the DSE-100 Series radiometer/photometers is provided on the Certificate of Limited Warranty enclosed separately with each unit.

NOTE: For assistance of any kind, including help with a radiometer under warranty, contact the Customer Service Department at Spectronics Corporation. In the U.S. and Canada, call toll-free 1-800-274-8888. Give full details of the difficulty and include the model and serial numbers of the unit and the date of purchase. If return of the readout unit or sensor(s) to the factory is deemed necessary, shipping instructions will be provided. If an estimate of charges for nonwarranty work or other service work is required, a quote will be furnished upon evaluation of the unit. Out-of-warranty service work will not be performed without customer approval.

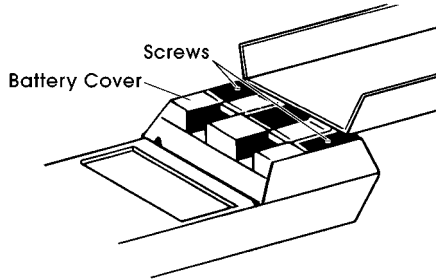
6.3 SHIPPING

Pack the radiometer in the original shipping materials and container. Insure the unit for full value and ship it prepaid to the factory.

6.4 BATTERY COVER REMOVAL

The battery cover at the back of the radiometer can be removed for maintenance purposes by the following procedure:

- Disconnect the charger (if used) from the unit.
- Turn the two screws in the battery cover counterclockwise until the cover is removed.
- Reverse this procedure when replacing the cover.

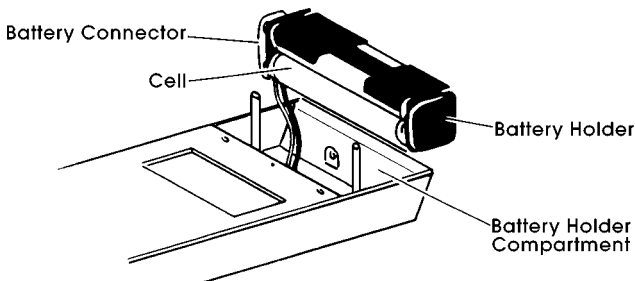


6.5 BATTERY INSTALLATION AND REPLACEMENT

The radiometer is provided with nonrechargeable, "AA" size alkaline cells as standard. If rechargeable nickel-cadmium batteries are used, the Spectrolite DM-15C0001 bivoltage AC recharger must be used. Use of any other charger may damage the meter. Refer to the markings on the DM-15C0001 for power source requirements.

Battery cells should be replaced as follows:

- Lift out the battery holder from the radiometer and remove the holder from the battery connector.
- Take out the old cells from the holder.
- Insert four new cells, ensuring that the cells are installed according to the polarity designated on the battery holder. **FAILURE TO DO SO CAN DAMAGE THE RADIOMETER.**
- Check that the battery holder contacts are clean and making good connection.
- Reconnect the battery holder to the battery connector.
- Replace the battery holder in its compartment and replace the battery cover.



6.6 BATTERY CHARGING

Nickel-cadmium batteries should be recharged as follows:

- a. Be sure the radiometer is turned off.
- b. Plug the jack of the DM-15C0001 recharger into the receptacle at the back of the radiometer.
- c. Set the recharger at the proper voltage and plug it into an appropriate AC power source.
- d. Approximately 12 hours are required to fully charge the batteries.

6.7 RECHARGEABLE BATTERY CARE

- a. Avoid discharging rechargeable batteries completely or leaving discharged cells in the radiometer for long periods of time.
- b. Ensure that all newly purchased rechargeable cells are charged for at least 15 minutes before operating the radiometer.

6.8 PREVENTIVE MAINTENANCE

6.8.1 Daily Care

- a. Immediately clean all spilled materials from the radiometer and wipe dry. If necessary, moisten a cloth with soap and water to clean plastic surfaces and the sensor head. The sensor surface should be rinsed with ethanol to remove any residual soap and/or organic contaminants.
- b. Whenever possible, avoid exposure or usage in areas that are subject to temperature and humidity extremes, vibration or mechanical shock, dust or corrosive fumes, or strong electrical or electromagnetic interference.



WARNING

Do not attempt to clean this instrument when it is connected to the charger.

6.8.2 Monthly Care

Verify unit calibration by performing operational checks using known value sources. If the need for recalibration is indicated, contact the factory.

6.8.3 Annual Care

It is recommended that the sensor(s) and readout unit be returned to the factory for a complete overall check and recalibration **at least every 12 months**. Contact the Customer Services Department for shipping instructions before returning the unit. In the U.S. and Canada, call 1-800-274-8888.

6.8.4 Storage

When the radiometer is not in use, store it in a location free from temperature extremes, dust or corrosive fumes, and vibration or mechanical shock.

6.9 TROUBLESHOOTING

6.9.1 If the radiometer does not yield satisfactory results, follow this procedure before attempting to service the unit:

- a. Review and comply with the Irradiance Measurement Procedure, listed in paragraph 4.2.
- b. Check that the irradiance intensity range being measured is within the rating of the unit.

- c. Be sure that the batteries are properly installed (refer to paragraph 6.5) and that they are charged.
- d. Ensure that the environment in which the radiometer is being used is within the unit specifications.
- e. Inspect the sensor head and readout unit for visible damage. If any damage is observed, the radiometer should be returned to the factory. Be sure to contact the Customer Service Department at Spectronics Corporation for shipping instructions *before* returning the radiometer. In the U.S. and Canada, call toll-free 1-800-274-8888. A quote will be furnished upon receipt and evaluation of the unit. Out-of-warranty service work will not be performed without customer approval.

6.9.2 If the steps taken in paragraph 6.9.1 do not yield satisfaction, refer to the Troubleshooting Chart (Table 6-1). Direct all other repair and adjustment needs to the factory.

TABLE 6-1 TROUBLESHOOTING

Symptom	Probable Cause	Cure
1. Low intensity on numerical display.	Batteries not fully charged, defective or making poor contact.	Remove battery cover and check that batteries are making good contact. Refer to paragraph 6.5. If good contact is made, check the voltage. If symptom persists, replace batteries.
2. No numerical display.	a. Batteries not installed.	a. Remove battery cover to ascertain whether batteries are present.
	b. Batteries installed incorrectly.	b. Remove batteries and check that polarity is correct.
	c. Batteries not making contact.	c. Ensure that good contact is made. Check voltage. If symptom persists, replace batteries.
	d. Battery voltage below desired level.	d. Recharge nicad batteries or replace alkaline batteries.
3. Display registers a zero reading when exposed to light source.	Sensor connector not in sensor receptacle properly.	Connect sensor to sensor receptacle correctly.
4. Green LED Indicator not lit.	Battery voltage below desired level.	Recharge nicad batteries or replace alkaline batteries.

SECTION VII ORDERING INFORMATION

TABLE 7-1

Description	Part No.
BATTERY, Alkaline, ``AA`` Size, 1.5 Volt (Four Required)	Commercial
BATTERY, Nicad Rechargeable, ``AA`` Size, 1.25 Volt (Four Required)	DM-N4
BATTERY RECHARGER, Bivoltage (120V/60Hz, 230V/50Hz)	DM-15C0001
CARRYING CASE	DR-75

ACCESSORY



DR-75 Carrying Case